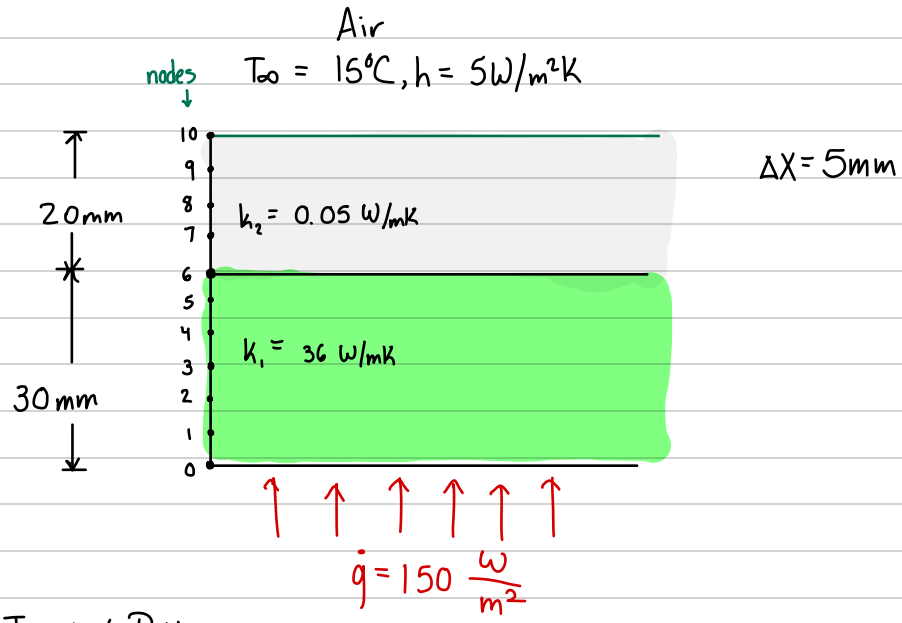
Problem 2

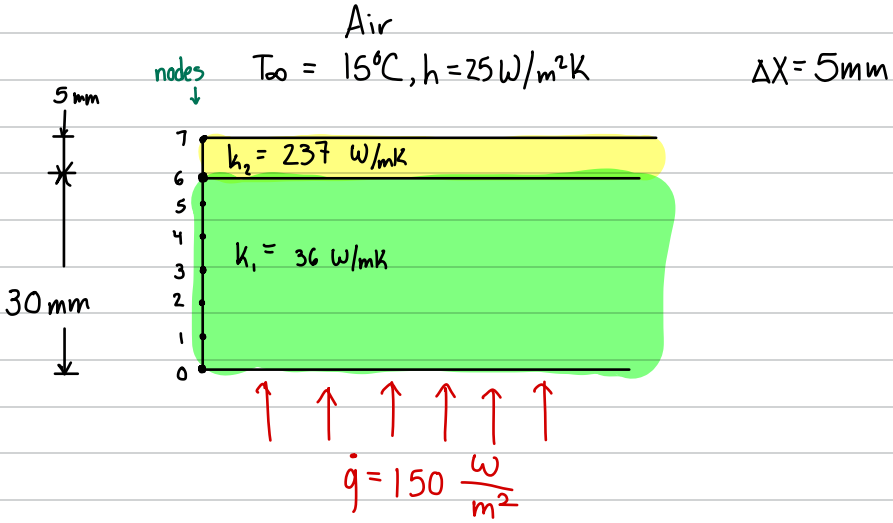
For comparison, 2 cases must be solved in transient mode.

Case A - Original Problem Made Transient



Case B: Adapted Transient Problem

For faster thermal response the nonmetal was replaced with a 5mm aluminum plate ($k = 237$ W/m K) and h was increased to 25 W/m²K (forced air), reducing the thermal resistance of SB-96 plate.



$$M = \frac{0.030 + 0.005}{0.005} + 1 = 8 \text{ nodes}$$

Assumptions:

- One dimensional - x
- Uniform properties:
- h = const, k = const, $\dot{q}$ = const
- Radiation is negligible
- Mechanisms: conduction and convection
- $T_i = T_\infty = 15°C$
- Perfect contact between plates
- Case A non metal approximated to cork.
- Case B top material replaced with aluminum.
- No heat generation
- 5% threshold - energy storage

Data:

Property	SB-96 CuSi (A & B)	A: Nonmetal	B: Aluminum
k (W/mK)	36	0.05	237
ρ (kg/m^3)	8530	86	2702
C_p (J/kgK)	380	2030	903
L (m)	0.030	0.020	0.005
h (W/m^{2}K)		5	25

*Values obtained from Cengel Textbook-Appendix 1

Procedure

1) Biot Number $Bi = \frac{hL_c}{k}$ $L_c = \frac{V}{A_s}$

Case A:

Non metal: $Bi = 5\frac{W}{m^2K} \cdot 0.020m \,/\, 0.05\frac{W}{mK} = 2$ $Bi > 0.1$

$\rho C_p L \rightarrow 86\frac{kg}{m^3} \cdot 2030\frac{J}{kgK} \cdot 0.020m = 3491.6\frac{J}{m^2K}$

SB-96 plate: $Bi = \frac{U_A L}{k} \Rightarrow Bi = 1.667\frac{W}{m^2K} \cdot 0.030m \,/\, 36\frac{W}{mK} = 0.001389$ $Bi < 0.1$

$R_{Total} = R_{cond} + R_{conv} = \frac{L_2}{k_2 A} + \frac{1}{h_2 A}$

$R_{Total} \cdot A = \frac{L_2}{k_2} + \frac{1}{h_2} = R_{TA}$

$R_{TA} = \frac{0.020}{0.05} + \frac{1}{5} = 0.6\frac{m^2K}{W}$

$U_A = \frac{1}{R_{Total}} = \frac{1}{0.6} = 1.667\frac{W}{m^2K}$

$\rho C_p L \rightarrow 8530\frac{kg}{m^3} \cdot 380\frac{J}{kgK} \cdot 0.030m = 97242\frac{J}{m^2K}$

$R_{TA} = R_{Total} A = \frac{L_2}{k_2} + \frac{1}{h}$

$\dot{Q} = \frac{\Delta T}{R_{Total}} = \frac{A\Delta T}{R_{TA}} \therefore \dot{q} = \frac{\dot{Q}}{A} = \frac{\Delta T}{R_{TA}}$

U_A is used because nonmetal stores 3.6% of the energy and acts as a thermal resistance in series with convection.

$\frac{3491.6}{97242} = 0.036$

Case B

Aluminum plate: $Bi = 25 \frac{W}{m^2K} \cdot 0.005m / 237 \frac{W}{mK} = 0.000527$ $Bi < 0.1$

$\rho c_p L \rightarrow 2702 \frac{kg}{m^3} \cdot 903 \frac{J}{kgK} \cdot 0.005m = 12199.5 \frac{J}{m^2K}$

SB-96: $Bi = \frac{U_B L}{k} = 24.9868 \frac{W}{m^2K} \cdot 0.030m / 36 \frac{W}{mK} = 0.020822$ $Bi < 0.1$

$R_{Total} = \frac{0.005m}{237 \frac{W}{mK}} + \frac{1}{25 \frac{W}{m^2K}} = 0.040021 \frac{m^2K}{W}$

$U_B = 24.9868 \frac{W}{m^2K}$

$\rho c_p L \rightarrow 8530 \frac{kg}{m^3} \cdot 380 \frac{J}{kgK} \cdot 0.030m = 97242 \frac{J}{m^2K}$

*$\rho c_p L$ of whole assembly $= 97242 \frac{J}{m^2K} + 12199.5 \frac{J}{m^2K} = 109442 \frac{J}{m^2K}$

Case A: $\frac{3491.6}{97242} = 0.036$ negligible

Case B: $12199.5/97242 = 0.125$ not negligible

Aluminum stores 12.5% of the SB-96 energy, above the ~5% threshold used in this problem.

2) $\dot{Q}_{conv} = hA_s (T_\infty - T)$

$\dot{Q} = \frac{T_\infty - T}{R_{Total}}$ $R_{Total} = \frac{L_2}{k_2 A_s} + \frac{1}{hA_s} = \frac{1}{A_s}\left(\frac{L_2}{k_2} + \frac{1}{h}\right)$

$\dot{Q} = \frac{T_\infty - T}{\frac{1}{A_s}\left(\frac{L_2}{k_2} + \frac{1}{h}\right)} = \frac{A_s (T_\infty - T)}{\frac{L_2}{k_2} + \frac{1}{h}}$

$\dot{Q} = U A_s (T_\infty - T)$ $\dot{Q} = \dot{q} A_s$

$[\dot{q}_0 A_s + U A_s (T_\infty - T)] dt = mc_p dT$

$mc_p = \sum \rho V c_p = \sum \rho A_s L c_p = A_s \sum \rho L c_p$

Case A: $A_s (\rho_1 L_1 c_{p_1} + \rho_2 L_2 c_{p_2})$ negligible: storage is only 3.6%

Case B: $A_s (\rho_1 L_1 c_{p_1} + \rho_2 L_2 c_{p_2})$

$$\left[\dot{q}_0 A_s + U A_s (T_\infty - T)\right] dt = A_s \left(\Sigma \rho L c_p\right) dT$$

$$\left[\dot{q}_0 + U(T_\infty - T)\right] dt = \left(\Sigma \rho c_p L\right) dT$$

$$q_0 + U(T_\infty - T) = -U(T - T_\infty) + q_0 = -U\left(T - T_\infty - \frac{\dot{q}_0}{U}\right)$$

θ

$$-U\left(T - T_\infty - \frac{\dot{q}_0}{U}\right) dt = \left(\Sigma \rho c_p L\right) dT$$

$$d\theta = dT - dT_\infty - d\left(\frac{\dot{q}_0}{U}\right)$$

(dT_∞ and $d\left(\frac{\dot{q}_0}{U}\right)$ are 0)

U, T_∞ and $\dot{q}_0$ are constants

$$-\frac{U\,\theta\, dt}{\theta} = \left(\Sigma \rho c_p L\right) \frac{d\theta}{\theta}$$

$$-U dt = \left(\Sigma \rho c_p L\right) \frac{d\theta}{\theta} \Rightarrow -\underbrace{\frac{U}{\Sigma \rho c_p L}}_{b} dt = \frac{d\theta}{\theta}$$

$$\int_0^t -b\, dt = \int_{\theta_i}^{\theta} \frac{d\theta}{\theta}$$

$$|\ln|\theta| - \ln|\theta_i|| = -b(t - 0)$$

$$\ln \frac{\theta}{\theta_i} = -bt \Rightarrow \frac{\theta}{\theta_i} = \frac{T(t) - T_\infty - \frac{\dot{q}_0}{U}}{T_i - T_\infty - \frac{\dot{q}_0}{U}} = e^{-bt}$$

$$\theta_i = T_i - T_\infty - \frac{\dot{q}_0}{U}$$

Since $T_i = T_\infty = 15°C$

$$\theta = \theta_i e^{-bt} \Rightarrow T - T_\infty - \frac{\dot{q}_0}{U} = -\frac{\dot{q}_0}{U} e^{-bt}$$

$$T = T_\infty + \frac{\dot{q}_0}{U} - \frac{\dot{q}_0}{U} e^{-bt}$$

$$T(t) = T_\infty + \frac{\dot{q}_0}{U}\left(1 - e^{-bt}\right)$$

Case A:

$$\Sigma \rho c_p L = 97\,242 \frac{J}{m^2 K}$$

$$b_A = \frac{1667 \frac{W}{m^2 K}}{\left(8530 \frac{kg}{m^3}\right)\left(380 \frac{J}{kg K}\right)(0.030 m)} = 1.7143 \cdot 10^{-5} 1/s$$

$$\frac{1}{b_A} = 58344 s \left(\frac{1 h}{3600 s}\right) = 16.2067 h$$

$$T_{ssA} = 15°C + \frac{150 W/m^2}{1.667 \frac{W}{m^2 °C}} = 105 °C$$

$$\frac{(T - T_\infty) U_A}{\dot{q}_0} = \frac{(93°C - 15°C)\left(1.667 \frac{W}{m^2 °C}\right)}{150 W/m^2} = 0.8667$$

$$\frac{\ln(1 - 0.8667)}{-1.7143 \cdot 10^{-5} 1/s} = t_{q3} = 117551 s = 32.6531 h$$ ← Time to reach T=93°C

$$\left.\frac{dT}{dt}\right|_0 = \frac{\dot{q}_0}{\Sigma \rho c L} = \frac{150 W/m^2}{97242 \frac{J}{m^2 K}} = 5.553 °C/h$$

$$\alpha_1 = \frac{k_1}{\rho_1 c_1} = \frac{36 \frac{W}{mK}}{8530 \frac{kg}{m^3} \cdot 380 \frac{J}{kg K}} = 1.111 \cdot 10^{-5} m^2/s$$

$$\tau = \frac{\alpha_1 t_{q3}}{L_c^2} = \frac{\left(1.111 \cdot 10^{-5} m^2/s\right)(117551 s)}{(0.030 m)^2} = 1450.6$$

$$Bi \cdot \tau = \left(1.389 \cdot 10^{-3}\right)(1450.6) = 2.015 \overset{\checkmark}{=} b_A t_{q3}$$

Case B:

$\Sigma \rho c_p L = 97242 + 12199.5 = 109442 \frac{J}{m^2 K}$

$b_B = \frac{24.99}{109442} = 2.283 \cdot 10^{-4} \frac{1}{s}$

$\frac{1}{b_B} = 4379.43\,s = 1.22\,h$

$T_{ssB} = 15°C + \frac{150 \frac{W}{m^2}}{24.99 \frac{W}{m^2 °C}} = 21.0024°C$

$t_{93} \Rightarrow \frac{(93-15)(24.99)}{150} = 12.99$

$\frac{\ln(1-12.99)}{-2.283 \cdot 10^{-4}}$ = no solution — argument is negative.

Plate does not reach 93°C

$\left.\frac{dT}{dt}\right|_0 = \frac{150}{109442} = 4.93\ °C/h$

Case B reaches steady state in less time than A since $b_B > b_A$ and

$\frac{b_B}{b_A} = \frac{2.283 \cdot 10^{-4}\ 1/s}{1.7143 \cdot 10^{-5}\ 1/s} = 13.32$

Note: Initial rate is slightly lower in Case B because aluminum adds 12.5% more thermal mass, but the larger U lets heat escape so B remains at 21°C within hours instead of rising towards 105°C.

Time constants

$1.22h_B$ vs $16.2067h_A$

$16.2067 / 1.22 = 13.3$ — How long the materials respond to change.

Explicit Finite Difference

$$\dot{Q} + \cancel{\dot{E}_{gen}}^{\,0} = \rho V c_p \frac{T_m^{i+1} - T_m^i}{\Delta t}$$

$m=0$

$$\frac{2\Delta x}{k_1 A}\left(\dot{q}_0 A + k_1 A \frac{T_1^i - T_0^i}{\Delta x} = \rho_1 A \frac{\Delta x}{2} c_1 \frac{T_0^{i+1} - T_0^i}{\Delta t}\right)$$

$$\frac{2\dot{q}_0 \Delta x}{k_1} + 2(T_1^i - T_0^i) = \underbrace{\frac{\rho_1 c_1}{k_1}}_{1/\alpha} \frac{(\Delta x)^2 (T_0^{i+1} - T_0^i)}{\Delta t} = \frac{T_0^{i+1} - T_0^i}{\tau_1} \qquad \tau = \frac{\alpha \Delta t}{\Delta x^2}$$

$$\frac{2 \dot{q}_0 \Delta x \tau_1}{k_1} + 2T_1^i \tau_1 - 2T_0^i \tau_1 + T_0^i = T_0^{i+1} \qquad (1)$$

Case A: $m = 1\text{-}5$ (SB-96) $m = 7\text{-}9$ (non metal)

$$\left(kA \frac{T_{m-1}^i - T_m^i}{\Delta x} + kA \frac{T_{m+1}^i - T_m^i}{\Delta x} = \rho A \Delta x c_p \frac{T_m^{i+1} - T_m^i}{\Delta t}\right) \Big/ \frac{kA}{\Delta x}$$

$$\frac{T_m^{i+1} - T_m^i}{\tau} = T_{m-1}^i - 2T_m^i + T_{m+1}^i$$

$$T_m^{i+1} = \tau\left(T_{m-1}^i + T_{m+1}^i\right) + (1 - 2\tau) T_m^i \qquad (2)$$

$m = 6$ interface

$$\left(k_1 A \frac{T_5^i - T_6^i}{\Delta x} + k_2 A \frac{T_7^i - T_6^i}{\Delta x} = (\rho_1 c_1 + \rho_2 c_2) A \frac{\Delta x}{2} \frac{T_6^{i+1} - T_6^i}{\Delta t}\right) \Big/ \frac{A}{2\Delta x}$$

$$2k_1 (T_5^i - T_6^i) + 2k_2 (T_7^i - T_6^i) = \frac{(\rho_1 c_1 + \rho_2 c_2) \Delta x^2 (T_6^{i+1} - T_6^i)}{\Delta t}$$

$$\beta = \frac{2\Delta t}{(\rho_1 c_1 + \rho_2 c_2)\Delta x^2}$$

$$T_6^{i+1} = \left[1 - \beta(k_1 + k_2)\right] T_6^i + \beta k_1 T_5^i + \beta k_2 T_7^i \qquad (3)$$

Top node Case A $m = 10$ & Case B $m = 7$

$$\left(hA(T_\infty - T_n^i) + k_2 A \frac{T_{n-1}^i - T_n^i}{\Delta x} = \rho_2 A \frac{\Delta x}{2} c_2 \frac{T_n^{i+1} - T_n^i}{\Delta t}\right) \Big/ \frac{k_2 A}{2\Delta x}$$

$$\frac{2h\Delta x}{k_2}(T_\infty - T_n^i) + 2(T_{n-1}^i - T_n^i) = \frac{T_n^{i+1} - T_n^i}{\tau_2}$$

$$T_n^{i+1} = \left(1 - 2\tau_2 - 2\tau_2 \frac{h\Delta x}{k_2}\right) T_n^i + 2\tau_2 T_{n-1}^i + 2\tau_2 \frac{h\Delta x}{k_2} T_\infty \qquad (4)$$

Eq. (1), (3), (4) are half volume nodes ($\Delta x/2$) hence why their equations carry a 2.

Primary Coefficients

For SB-96 (1), (2)

$$1 - 2r_1 \geq 0 \qquad \Delta t \leq \frac{\Delta x^2}{2\alpha_1} = \frac{(0.005m)^2}{2 \cdot 1.111 \cdot 10^{-5} \frac{m^2}{s}} = 1.125\,s$$

Interface (3)

$$1 - \beta(k_1 + k_2) \geq 0 \rightarrow \Delta t \leq \frac{(\rho_1 c_1 + \rho_2 c_2)\Delta x^2}{2(k_1 + k_2)}$$

Case A: $$\frac{\left(8530 \frac{kg}{m^3} \cdot 380 \frac{J}{kgK} + 86 \frac{kg}{m^3} \cdot 2030 \frac{J}{kgK}\right)(0.005m)^2}{2\left(36 \frac{W}{mK} + 0.05 \frac{W}{mK}\right)} = 1.1845\,s$$

Case B: $$\frac{\left(8530 \frac{kg}{m^3} \cdot 380 \frac{J}{kgK} + 2702 \frac{kg}{m^3} \cdot 903 \frac{J}{kgK}\right)(0.005m)^2}{2\left(36 \frac{W}{mK} + 237 \frac{W}{mK}\right)} = 0.2601\,s$$

Non metal (2)

$$\Delta t \leq \frac{(0.005m)^2}{2(2.864 \cdot 10^{-7} \frac{m^2}{s})} = 43.65s \qquad \alpha_2 = \frac{k_2}{\rho_2 c p_2} = \frac{0.05 \frac{W}{mK}}{(86 \cdot 2030) \frac{J}{m^3 K}} = 2.864 \cdot 10^{-7} \frac{m^2}{s}$$

Surface (4)

$$1 - 2r_2\left(1 + h\Delta x / k_2\right) \geq 0 \rightarrow \Delta t \leq \frac{\Delta x^2}{2\alpha_2(1 + h\Delta x / k_2)}$$

Case A: $$\frac{h\Delta x}{k_2} = \frac{5 \frac{W}{m^2 K} \cdot 0.005m}{0.05 \frac{W}{mK}} = 0.5 \Rightarrow \frac{(0.005m)^2}{2\left(2.864 \cdot 10^{-7} \frac{m^2}{s}\right)\left(1 + \frac{5W}{m^2K}(0.005m)/0.05 \frac{W}{mK}\right)} = 29.097\,s$$

Case B: $$\frac{25 \frac{W}{m^2 K} \cdot 0.005m}{237 \frac{W}{mK}} = 5.27 \cdot 10^{-4} \Rightarrow \frac{(0.005m)^2}{2(9.7 \cdot 10^{-5} \frac{m^2}{s})\left(1 + \frac{(25 W/m^2K) \cdot (0.005m)}{237 W/mK}\right)} = 0.1286\,s$$

$$\alpha_2 = \frac{k_2}{\rho_2 c p_2} = \frac{237 \frac{W}{mK}}{(2702 \cdot 903) \frac{J}{m^3 K}} = 9.7 \cdot 10^{-5} \frac{m^2}{s}$$

Case A: $\Delta t_{max} = 1.125s$ $\Delta t = 1s$

Case B: $\Delta t_{max} = 0.1286s$ $\Delta t = 0.1s$

Chosen due to being the most restrictive limit of each case.

Case A

$$\tau_1 = \frac{\left(1.111\cdot10^{-5}\ \frac{m^2}{s}\right)(1s)}{(0.005m)^2} = 0.4444 \qquad \therefore\ 1-2\tau_1 = 0.1112 \qquad 2\tau_1 < 0.8888$$

$$2\tau_1 \frac{\dot{q}_0 \Delta x}{k_1} = \frac{2(.4444)\left(150\ \frac{W}{m^2}\right)(0.005m)}{36\ W/mK} = 0.01852$$

$$\beta = 2(1s) \Big/ \left[\left(8530\ \frac{kg}{m^3}\cdot 380\ \frac{J}{kgK} + 86\ \frac{kg}{m^3}\cdot 2030\ \frac{J}{kgK}\right)(0.005m)^2\right] = 0.023419\ \frac{mK}{W}$$

$$\beta k_1 = 0.843096\ \frac{mK}{W} \qquad \beta k_2 = 0.001171\ \frac{mK}{W}$$

$$1-\beta(\Sigma k_i) = 0.155733$$

$$\tau_2 = \frac{2.864\cdot10^{-7}\ \frac{m^2}{s}\ (1s)}{(0.005m)^2} = 0.011456$$

$$1-2\tau_2-2\tau_2\left(\frac{h\Delta x}{k_2}\right) = 0.965632$$

$$2\tau_2 = 0.022912$$

$$1-2\tau_2 = 0.977088$$

$$T_0^{i+1} = 0.01852 + 0.8888\ T_1^i + 0.1112\ T_0^i$$

$$T_m^{i+1} = 0.4444\left(T_{m-1}^i + T_{m+1}^i\right) + 0.1112\ T_m^i \qquad \text{for} \quad m=1-5$$

$$T_6^{i+1} = 0.155733\ T_6^i + 0.843096\ T_5^i + 0.001171\ T_7^i$$

$$T_m^{i+1} = 0.011456\left(T_{m-1}^i + T_{m+1}^i\right) + 0.977088\ T_m^i \qquad \text{for} \quad m=7-9$$

$$T_{10}^{i+1} = 0.965632\ T_{10}^i + 0.022912\ T_9^i + 0.17184$$

Case B:

Similar to Case A, but remembering that $h = 25 \frac{W}{m^2 K}$, $k_2 = 237 \frac{W}{mK}$, $\Delta t = 0.1s$

$$\tau_1 = \frac{(1.111 \cdot 10^{-5} m^2/s \cdot 0.1s)}{(0.005m)^2} = 0.04444$$

$$\beta = \frac{2(0.1s)}{\left(8530 \frac{kg}{m^3} \cdot 380 \frac{J}{kgK} + 2702 \frac{kg}{m^3} \cdot 903 \frac{J}{kgK}\right)(0.005m)^2} = 0.0014081$$

$$\tau_2 = \frac{(9.71 \cdot 10^{-5} \frac{m^2}{s})(0.1s)}{(0.005m)^2} = 0.3884$$

Used formulas from previous pages for calculations.

$$T_0^{i+1} = 0.001852 + 0.08888\, T_1^i + 0.91112\, T_0^i$$

$$T_m^{i+1} = 0.04444(T_{m-1}^i + T_{m+1}^i) + 91112\, T_m^i \quad \text{for} \quad m = 1-5$$

$$T_6^{i+1} = 0.615582\, T_6^i + 0.050692\, T_5^i + 0.333726 T_7^i$$

$$T_7^{i+1} = 0.22251\, T_7^i + 0.777\, T_6^i + 0.00615$$

Marching in time

Initial condition:

$T_m^0 = T_1 = 15°C$ for all nodes at $t=0$ $(i=0)$

Case A

$i=1$ $(t=\Delta t = 1s)$

$T_0^1 = 0.01852 + 0.8888(15) + 0.1112(15) = 15.01852\ °C$

$T_1^1 = 0.4444(15+15) + 0.1112(15) = 15.00\ °C$

$T_2^1 = \ldots = T_{10}^1 = 15.00°C$

$i=2$ $(t=2\Delta t = 2s)$

$T_0^2 = 0.01852 + 0.8888(15) + 0.1112(15.01852) = 15.020578°C$

$T_1^2 = 0.4444(15+15.01852) + 0.1112(15) = 15.008230°C$

$T_2^2 = \ldots = T_{10}^2 = 15.00°C$

Equation (1) - (4) are applied continuously using the temperatures of step i to obtain the ones from i+1.
Many steps are needed since the heat front advances one node at the time.

Case A - With Excel

i	t (s)	T0 (°C)	T1 (°C)	T2 (°C)	T3 (°C)	T4 (°C)	T5 (°C)	T6 (°C)	T7 (°C)	T8 (°C)	T9 (°C)	T10 (°C)
0	0	15	15	15	15	15	15	15	15	15	15	15
1	1	15.01852	15	15	15	15	15	15	15	15	15	15
2	2	15.02057942	15.00823029	15	15	15	15	15	15	15	15	15
3	3	15.02812351	15.0100607	15.00365754	15	15	15	15	15	15	15	15
4	4	15.03058929	15.01524225	15.0048777	15.00162541	15	15	15	15	15	15	15
5	5	15.03546884	15.01745647	15.00803839	15.00234839	15.00072233	15	15	15	15	15	15
6	6	15.03797944	15.02127577	15.00969515	15.00415441	15.00112395	15.000321	15	15	15	15	15
7	7	15.04165322	15.02355245	15.01237927	15.00526998	15.00211386	15.00053518	15.00027064	15	15	15	15
8	8	15.04408526	15.02663107	15.01418526	15.00702677	15.00281487	15.00111918	15.00049335	15.0000031	15	15	15
9	9	15.04709198	15.0288568	15.01653494	15.00833624	15.00393307	15.00159463	15.00102041	15.00000868	15.00000004	15	15
10	10	15.04940455	15.03148468	15.01836727	15.01002298	15.00485063	15.00237865	15.00150335	15.00002017	15.00000013	15	15

Longer times (Excel)

t (h)	T0	T1	T2	T3	T4	T5	T6	T7	T8	T9	T10
1	20.35	20.33	20.31	20.30	20.29	20.28	20.28	19.22	18.26	17.39	16.58
4	34.45	34.43	34.41	34.40	34.39	34.38	34.37	30.99	27.70	24.48	21.31
8	49.67	49.65	49.64	49.62	49.61	49.60	49.59	43.70	37.89	32.13	26.41
16	70.99	70.97	70.95	70.93	70.92	70.90	70.89	61.50	52.16	42.84	33.56
24	84.12	84.10	84.08	84.06	84.04	84.02	84.01	72.46	60.94	49.44	37.96
33	92.96	92.94	92.92	92.90	92.88	92.86	92.84	79.84	66.86	53.88	40.92
48	100.23	100.21	100.19	100.17	100.15	100.13	100.11	85.92	71.73	57.54	43.36
72	103.99	103.97	103.95	103.93	103.91	103.89	103.87	89.06	74.24	59.43	44.62
120	105.08	105.06	105.04	105.01	104.99	104.97	104.95	89.96	74.97	59.98	44.98
Steady ∞	105.125	105.10	105.08	105.06	105.04	105.02	105.00	90.00	75.00	60.00	45.00

Interpolation

t	T_0
33	92.96
x	93
48	100.23

$x = 33.0825$

93°C is reached when $t = 33.0825\ h$ similar to the lumped system $(t = 32.65h)$; a 1.3% difference.

Case B: *Similar procedure as Case A

$i=1$ ($t=0.1s$)

$T_0' = 0.001852 + 0.08888(15) + 0.91112(15) = 15.001852\,°C$

$T_1' = 0.04444(15+15) + 91112(15) = 15°C$

$T_m' = 15°C$ for $m = 1-7$

$T_0^2 = 0.001852 + 0.08888(15) + 0.91112(15.001852) = 15.003539\,°C$

$T_1^2 = 0.04444\,(15.001852+15) + 0.91112(15) = 15.000082\,°C$

T_2^2 to $T_7^2 = 15.00°C$

Equation (1) - (4) are applied continuously using the temperatures of step i to obtain the ones from i+1.
Many steps are needed since the heat front advances one node at the time.

Case B with Excel:

i	t (s)	T0 (°C)	T1 (°C)	T2 (°C)	T3 (°C)	T4 (°C)	T5 (°C)	T6 (°C)	T7 (°C)
0	0	15.00000000	15.00000000	15.00000000	15.00000000	15.00000000	15.00000000	15.00000000	15.00000000
1	0.1	15.00185200	15.00000000	15.00000000	15.00000000	15.00000000	15.00000000	15.00000000	15.00000000
2	0.2	15.00353939	15.00008230	15.00000000	15.00000000	15.00000000	15.00000000	15.00000000	15.00000000
3	0.3	15.00508413	15.00023228	15.00000366	15.00000000	15.00000000	15.00000000	15.00000000	15.00000000
4	0.4	15.00650490	15.00043773	15.00001365	15.00000016	15.00000000	15.00000000	15.00000000	15.00000000
5	0.5	15.00781765	15.00068851	15.00003190	15.00000075	15.00000001	15.00000000	15.00000000	15.00000000
6	0.6	15.00903601	15.00097615	15.00005970	15.00000211	15.00000004	15.00000000	15.00000000	15.00000000
7	0.7	15.01017165	15.00129360	15.00009787	15.00000457	15.00000013	15.00000000	15.00000000	15.00000000
8	0.8	15.01123457	15.00163501	15.00014686	15.00000852	15.00000032	15.00000001	15.00000000	15.00000000
9	0.9	15.01223336	15.00199548	15.00020684	15.00001431	15.00000067	15.00000002	15.00000000	15.00000000
10	1	15.01317542	15.00237096	15.00027777	15.00002226	15.00000125	15.00000005	15.00000000	15.00000000

t	T0	T1	T2	T3	T4	T5	T6	T7
1 min	15.129	15.109	15.093	15.080	15.070	15.063	15.059	15.059
10 min	15.821	15.802	15.785	15.771	15.760	15.751	15.745	15.745
30 min	17.088	17.068	17.050	17.035	17.021	17.009	17.000	16.999
1 h	18.444	18.424	18.405	18.387	18.371	18.357	18.343	18.342
2 h	19.943	19.922	19.902	19.883	19.864	19.846	19.829	19.826
3 h	20.605	20.584	20.563	20.543	20.523	20.504	20.484	20.481
4 h	20.896	20.876	20.855	20.835	20.814	20.794	20.774	20.771
6 h	21.082	21.061	21.041	21.020	20.999	20.978	20.958	20.954
10 h	21.125	21.105	21.084	21.063	21.042	21.021	21.000	20.997
Steady ∞	21.128	21.107	21.086	21.066	21.045	21.024	21.003	21.000

Lumped system $T_0 = 21.0024°C$ which approximates 21.128°C from the nodal method. The difference is due to the conduction drop across the SB-96 material. The lumped model ignores this.
Never reaches 93°C as expected.

Cooling - Heater off $\dot{q}_0 = 0$ $\quad 90°C \rightarrow 60°C$

Lumped

$$\frac{T(t) - T_\infty}{T_i - T_\infty} = e^{-bt}$$

$$t = \frac{1}{b} \ln \frac{T_i - T_\infty}{T - T_\infty} \Rightarrow bt = \ln \frac{90-15}{60-15} = 0.5108$$

$$t_A = \frac{0.5108}{1.7143 \cdot 10^{-5} 1/s} = 29797.9\,s = 8.2772\,h$$

$$t_B = \frac{0.5108}{2.283 \cdot 10^{-4} \frac{1}{s}} = 2237.41\,s = 37.2901\,min = 0.6215\,h$$

$$\frac{t_A}{t_B} = \frac{8.2772\,h}{0.6215\,h} = 13.32$$

Nodal

Heater off ($\dot{q}=0$), nodes start at 90°C and plate cools to $T_\infty = 15°C$ (reason why Case B can reach 60°C)

Case A: $T_0^{i+1} = 0.1112\,T_0^i + 0.8888\,T_1^i + 0.01852$

Case B: $T_0^{i+1} = 0.91112\,T_0^i + 0.08888\,T_1^i + 0.001852$

} Only for equation (1)

New marching is done considering $\dot{q}_0 = 0$, then we look for t when $T_0 = 60°C$

Case A

t	T0	T6 (interface)	T10 (surface)
0	90.00	90.00	90.00
10 min	89.68	89.64	42.78
1 h	86.05	86.00	38.82
2 h	81.87	81.83	37.42
4 h	74.24	74.20	34.86
6 h	67.47	67.44	32.59
8 h	61.48	61.45	30.58
8.53 h	60.00	59.97	30.09

Case B

t	T0	T6 (interface)	T7 (surface)
0	90.00	90.00	90.00
1 min	89.27	88.59	88.55
5 min	85.33	84.68	84.64
10 min	80.69	80.09	80.06
20 min	72.32	71.79	71.76
30 min	65.01	64.55	64.53
35 min	61.72	61.29	61.26
37.7 min	60.00	59.59	59.56

$$\frac{t_A}{t_B} = \frac{8.53h}{\left(\frac{37.7}{60}\,h\right)} = 13.58$$

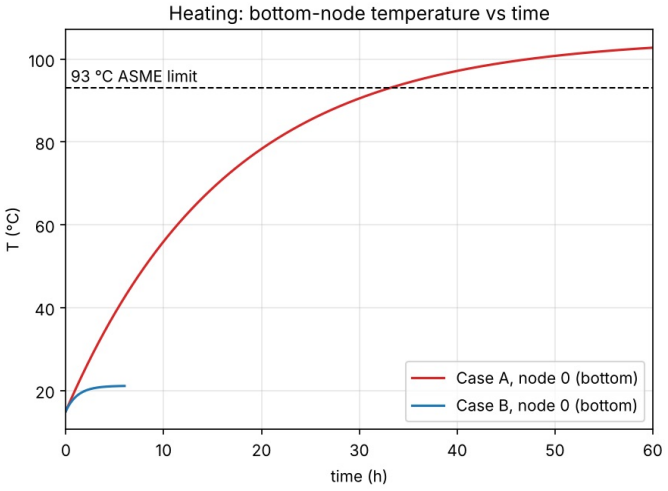
Ratios $\quad \%e = \frac{|13.32 - 13.58|}{13.58} = 1.915\%$

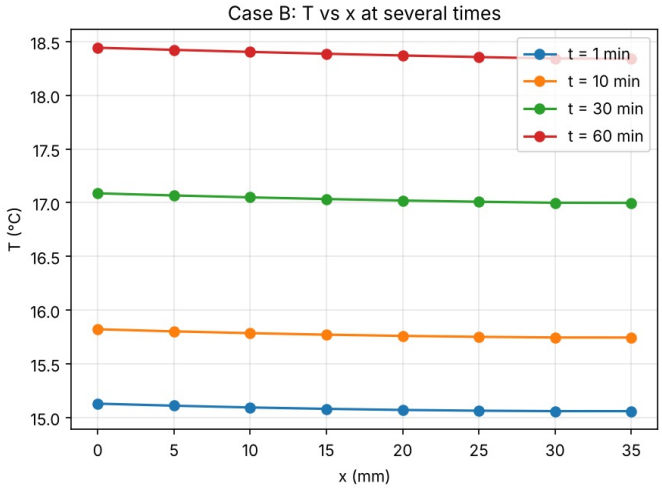
$t_A \quad \%e = 2.964\%$

$t_B \quad \%e = 1.088\%$

Conclusion

Adaptation increases U from 1.667 to 24.99 W/m²K while adding only 12.5% thermal mass. As a result, the time constant drops from 16.2h to 1.22 h (13.3X shorter) and the plate cools from 90°C to 60°C ~136 X faster (8.53h vs 37.7min). It also stays at 21°C keeping the SB-96 plate in compliance with the 93°C ASME limit which case A violates after 33h.





Claude generated the plots with our calculated data.